Project Proposal: *Purrsist! – Habit Tracking with Friends*

1. Project Overview

Purrsist! is a web-based habit tracking application designed to motivate users to build positive habits through **gamification**. Users track daily habits (e.g., studying, exercising, reading), and completing habits rewards them with **virtual pet cats** they can collect, evolve, and display.

The system combines **personal productivity tools** with **social interaction**, allowing users to view other users' achievements, cat collections, and progress. This encourages accountability and friendly competition.

The application will be built using a **client-server architecture** with a Flask backend and a JavaScript/HTML/CSS frontend.

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2. Problem and Motivation

Maintaining consistent habits is difficult because:

- Many habit trackers feel repetitive and unengaging.
- Users lack motivation to continue tracking habits long-term.
- Habit trackers often lack social features that encourage accountability.

Purrsist! addresses these issues by:

- Turning habit completion into a **reward system**
- Allowing users to **collect and interact with virtual pets**
- Providing **community interaction through shared progress**

This transforms habit tracking into a **fun and engaging experience** rather than a chore.

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3. Target Users

The application targets:

- **University students** who want to track study habits
- **People building routines** (exercise, reading, productivity)
- **Gamification enthusiasts**
- **Users who enjoy collecting virtual pets**

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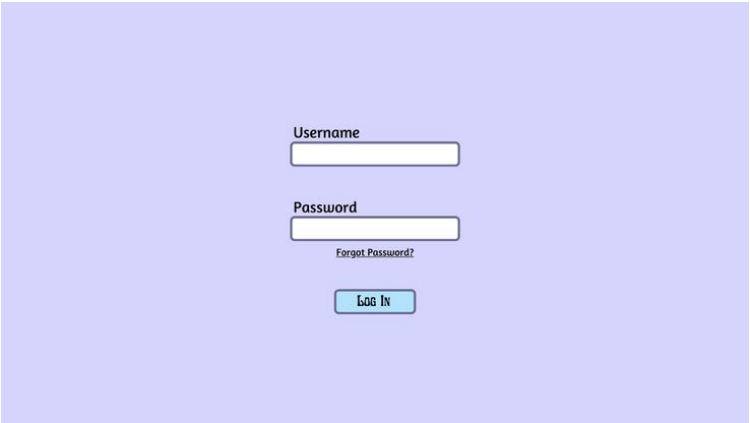
4. Core Features

4.1 User Authentication



Users must create an account to use the application.

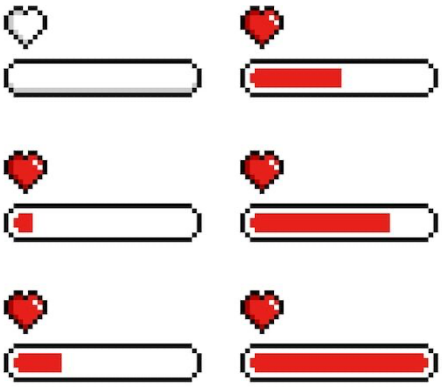
Features include:

- User registration
- Login
- Logout
- Secure password storage
- User session management

This satisfies the requirement that **users must be able to login and logout**.

4.2 Habit Tracking System

Users can create and manage habits they want to maintain.



Users can:

- Create habits (e.g., "Study 2 hours")
- Mark habits as **completed each day**
- Track completion history
- View habit streaks
- See statistics on habit consistency

Example habit data stored:

Habit	Frequency	Streak	Completion
Study	Daily	5 days	80%
Exercise	3x weekly	2 weeks	65%

This provides **value to users by giving insight into their behaviour**.

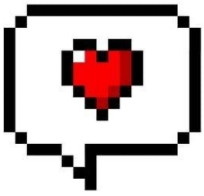
4.3 Virtual Cat Reward System

Gamification is the core engagement mechanic.

When users complete habits they earn:

- Affection points
- Cat treat rewards
- Adoption tokens
- New collectible cats

Cats will have:



- Names
- Rarity levels
- Happiness level
- Evolution stages

Example:

Cat Name	Rarity	Unlock Condition
Luna	Common	3 habit completions
Mochi	Rare	7 day streak
Astro	Epic	30 completions

Users can **view their cat collection in a "Cat Shelter" page**.

4.4 Pet Care and Wellbeing System

To reinforce the habit-building goal of the application, Purrsist! will include a **pet wellbeing mechanic** that links user consistency with the health and happiness of their virtual cats.

Each cat will have a **mood or wellbeing meter** that changes depending on how regularly the user completes their habits.

Behaviour rules

User Behaviour	Cat Response
Consistent habit completion	Cat remains happy
1–2 missed days	Cat mood slightly decreases
Several missed days	Cat becomes sad or inactive
Long inactivity	Cat temporarily leaves the shelter

Users can restore the cat's happiness by **resuming consistent habit completion**.

Purpose

This feature:

- Reinforces the importance of **consistent routines**
- Makes the virtual pet feel **alive and responsive**
- Adds emotional motivation without overly punishing the user

This system integrates with the reward mechanics and helps maintain engagement with the application.

4.5 Social Features

The project requires **users to view data from other users**.

Purrsist! will implement this through:

Public User Profiles

Each user has a profile showing:

- Habit streaks
- Cat collection
- Achievements

Leaderboard

Users can see:

- Longest streaks
- Most habits completed
- Largest cat collections

Friend Activity Feed (Optional stretch goal)

Displays:

- Friends completing habits
- Cats unlocked

This encourages friendly competition and community interaction.

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5. Application Architecture

The application will follow a **client-server architecture**:

```
Client (Browser)
    |
    | HTTP Requests
    |
Flask Web Server
    |
    | SQLAlchemy
    |
SQLite Database
```

Client Side

Technologies:

- HTML
- CSS
- Bootstrap or Tailwind
- JavaScript
- JQuery
- AJAX

Responsibilities:

- User interface
- Form submission
- Dynamic updates (AJAX)
- Habit progress display

Server Side

Technologies:

- Flask
- SQLAlchemy
- Flask-Login

Responsibilities:

- Authentication

- Business logic
- Habit tracking
- Reward generation
- API endpoints

Database

SQLite will store persistent data.

Key tables:

Users

id username password_hash

Habits

| id | user_id | habit_name | frequency |

Habit_Completions

| id | habit_id | date |

Cats

| id | name | rarity |

User_Cats

| id | user_id | cat_id |

This ensures **data persistence across sessions**.

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6. UI / UX Design Goals

The interface will be designed to be:

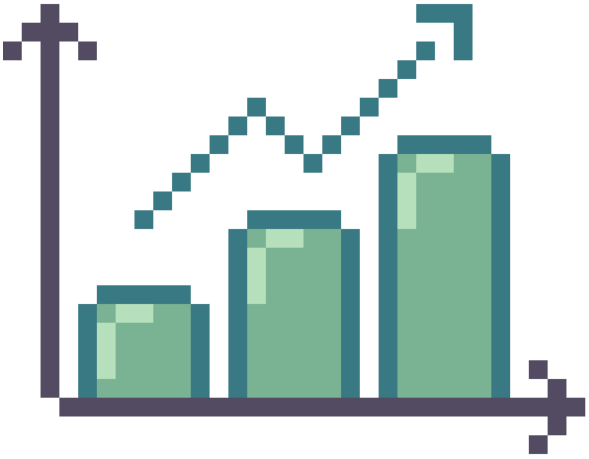
Engaging

- Cute cat themed UI
- Animations when unlocking cats
- Progress bars and rewards

Effective

- Clear dashboards
- Habit statistics
- Visible streaks

Intuitive



- Simple habit completion buttons
- Minimal navigation complexity
- Clean mobile-friendly layout

Example pages:

- Login / Register
- Dashboard
- Habit Manager
- Cat Collection
- Leaderboard
- Profile Page

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7. Example User Flow

Step 1: Register

User creates account.

Step 2: Create Habits

Example:

- Study
- Exercise
- Drink Water

Step 3: Complete Habit

User clicks **Complete Habit** button.

Step 4: Earn Rewards

System awards:

- XP
- Chance to unlock a cat

Step 5: View Collection

User visits **Cat Shelter** page to see collected cats.

Step 6: Compare with Others

User checks leaderboard or profiles.

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8. Agile Development Plan

The project will use **Agile methodology**.

Development will be divided into weekly milestones.

Week 1

Planning and setup

- GitHub repo
- Flask setup
- Database schema
- Authentication

Week 2

Core functionality

- Habit tracking
- Habit database
- Dashboard

Week 3

Gamification

- Cat collection system
- Reward mechanics

Week 4

Social features

- User profiles
- Leaderboards
- Viewing other users

Week 5

Polish and testing

- UI improvements
- Bug fixes
- Testing

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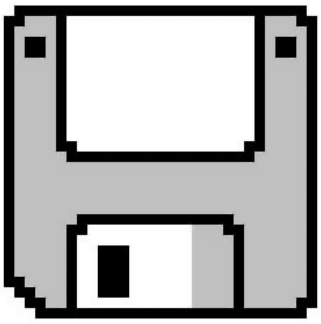
9. GitHub Collaboration Plan

Each team member will work on different components.

Example roles:

Member	Responsibility
Member 1	Authentication + database
Member 2	Habit tracking system
Member 3	Cat reward system
Member 4	UI/UX and social features



GitHub workflow:

- Feature branches
- Pull requests
- Code reviews

This ensures **proper collaboration and version control**.

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10. Testing Strategy

Testing will include:

Unit Tests

- Habit creation
- Reward generation
- Database operations

Integration Tests

- Login and session flow
- Habit completion pipeline

UI Testing

- Form validation
- Page navigation

Tests will run using **Python testing frameworks** such as pytest.

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11. Example Tech Stack Summary

Layer	Technology
Frontend	HTML, CSS, Bootstrap
Client Logic	JavaScript, JQuery
Server	Flask
Database	SQLite
ORM	SQLAlchemy
Async	AJAX
Authentication	Flask-Login

All technologies must follow the **assignment restrictions**.

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12. Key Deliverables

The GitHub repository will contain:

- Application source code
- README
- Database models
- Test suite
- Setup instructions
- Project documentation

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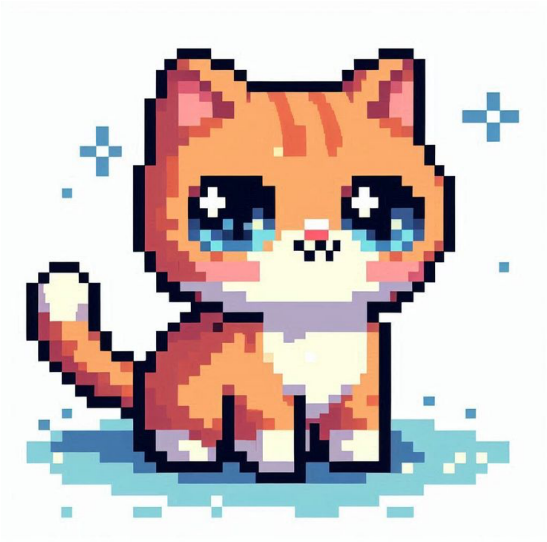
13. Project strengths

Purrsist! meets all technical requirements:

Requirement	Implementation
Client-Server Architecture	Flask + Browser
User Login / Logout	Flask-Login
Persistent Data	SQLite
View Other Users	Profiles + Leaderboards
Modern Web Technologies	HTML, CSS, JS, AJAX

It also scores highly on the design criteria:

- **Engaging:** gamification and collectibles
- **Effective:** meaningful habit analytics
- **Intuitive:** simple habit completion workflow